

IVF

In vitro fertilisation (IVF) is a process of [fertilisation](#) in which an [egg](#) is combined with [sperm in vitro](#) ("in glass").

Vitro - *In vitro* ([Latin](#) for "in glass"; often not italicized in English usage^{[3][4][5]}) studies are conducted using components of an organism that have been isolated from their usual biological surroundings. As the name suggests, *in vitro* experiments, colloquially "test-tube experiments", are traditionally done in glass labware, using test tubes, flasks, Petri dishes, etc.

Process:

The process involves monitoring and stimulating the [ovulatory process](#), then removing an ovum or ova (egg or eggs) from the [ovaries](#) and enabling sperm to fertilise them in a culture medium in a laboratory. After a fertilised egg ([zygote](#)) undergoes [embryo culture](#) for 2–6 days, it is [transferred](#) by [catheter](#) into the [uterus](#), with the intention of establishing a successful [pregnancy](#).

Embryo culture - is a component of [in vitro fertilisation](#) where in resultant embryos are allowed to grow for some time in an artificial medium.

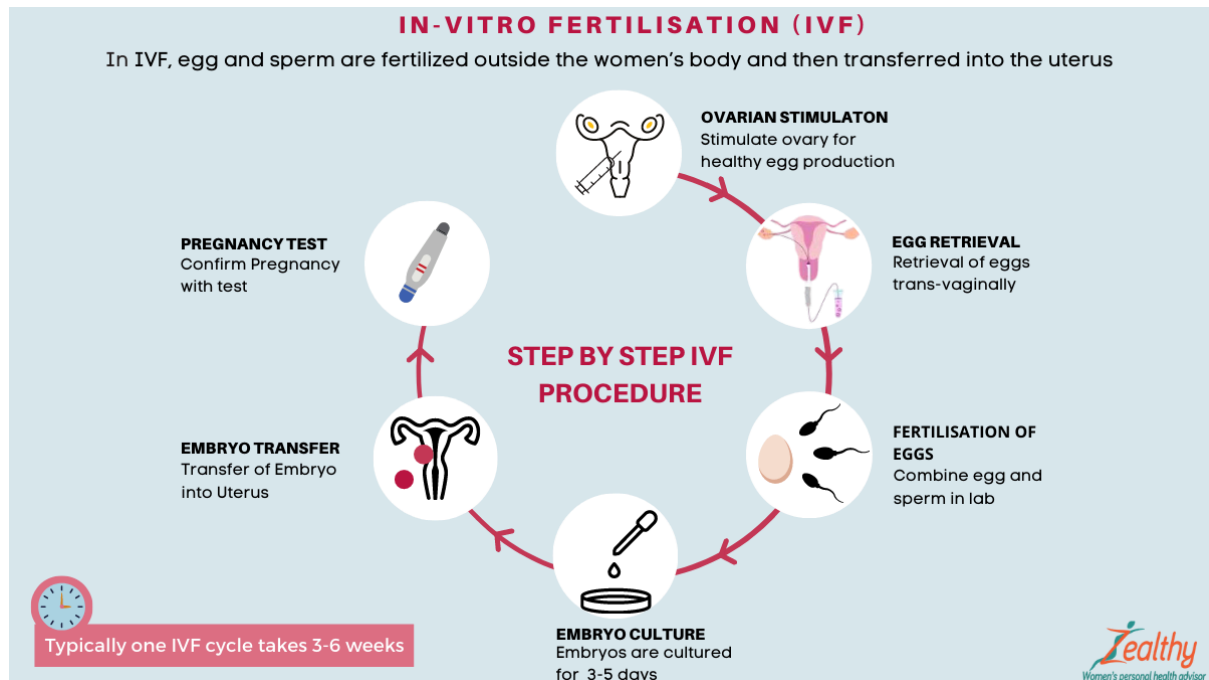
Assisted reproductive technology (ART) includes medical procedures used primarily to address [infertility](#).

An **oocyte** (/ˈoʊəsaɪt/, **oöcyte**, or **ovocyte**) is a female [germ cell](#) involved in [sexual reproduction](#). An oocyte is an [immature ovum](#), an immature [egg cell](#) produced in a female fetus in the [ovary](#) during [gametogenesis](#). During [oogenesis](#), the oogonia become primary oocytes. An oocyte is a form of genetic material that can be collected for [cryopreservation](#).

When assisted by [egg donation](#) and IVF, many women who have reached [menopause](#), have infertile partners, or have idiopathic female-fertility issues, can still become pregnant. After the IVF treatment, some couples get pregnant without any fertility treatments

Process

IVF Process Overview



In Vitro Fertilization (IVF) is a fertility treatment in which eggs are collected from the ovaries, fertilized with sperm in a laboratory, and the resulting embryo is transferred into the uterus.

1. Preparation

Before starting IVF, the patient and partner undergo a **medical history review, fertility testing, and relevant investigations**. Medications such as birth control pills or estrogen may be used to control cycle timing and prepare for treatment.

2. Ovarian Stimulation

Hormonal injections are given to stimulate the ovaries to develop **multiple mature eggs instead of the single egg normally produced during a cycle**. Egg development is monitored through scans and/or blood tests. Once the follicles are ready, a **trigger injection** is given, typically about **36 hours before egg retrieval**, to complete egg maturation.

3. Egg Retrieval

Mature eggs are collected using a **thin needle passed through the vagina into the ovarian follicles**, usually with sedation or medication for comfort. The retrieved eggs are placed in a laboratory culture medium and transferred to an incubator.

4. Fertilization

The retrieved eggs are fertilized with sperm in the laboratory. **ICSI (Intracytoplasmic Sperm Injection)** may be used, where a single sperm is injected directly into each mature egg. On average, around **70% of mature eggs may fertilize**.

5. Embryo Development

Fertilized eggs are monitored as they divide and develop over the next several days. Approximately **half of fertilized embryos may reach the blastocyst stage**, when they may be suitable for transfer or freezing. Suitable embryos can be transferred immediately or **cryopreserved for future use**.

6. Embryo Transfer

A selected embryo is transferred into the uterus using a **thin catheter passed through the cervix**. The procedure is usually short and similar to a pelvic examination. Mild **cramping, bloating, breast tenderness, spotting, or constipation** may occur afterward.

7. Pregnancy Confirmation

Approximately **14 days after embryo transfer**, a blood pregnancy test is performed. If positive, the patient proceeds to prenatal care. If negative, another frozen embryo transfer may be considered if embryos remain; otherwise, another IVF cycle may be undertaken.

Key Takeaways

- IVF involves **stimulation → egg retrieval → fertilization → embryo development → embryo transfer → pregnancy testing**.
- The number of eggs retrieved does not equal the number of embryos available for transfer; attrition occurs at each stage.
- **Age at egg retrieval** is an important factor influencing IVF success rates.
- Additional suitable embryos can be **frozen for future transfers**.

Pre-IVF Testing (Detailed Summary)

This report describes the **diagnostic and evaluation stage before starting an IVF cycle**. The purpose is to understand both partners' reproductive health, identify factors that could affect fertilization or pregnancy, and use the findings to create a **personalized IVF treatment plan**.

Test / Evaluation	Applicable For	Stage	Why is it done?	What information does it reveal / Purpose
Medical & Fertility History	Both partners	Before IVF	To understand previous conception attempts, medical conditions, and fertility history	Identifies potential causes/risk factors for infertility and guides further evaluation
AMH	Female	Pre-IVF evaluation	To assess ovarian reserve	Indicates ovarian reserve and helps with IVF stimulation planning
FSH	Female	Pre-IVF evaluation	To assess ovarian function	Provides information about ovarian reserve and reproductive hormone function
LH	Female	Pre-IVF evaluation	To assess reproductive hormonal function	Provides information about hormonal balance and ovulatory function
Prolactin	Female	Pre-IVF evaluation	To identify hormonal abnormalities	Abnormal levels may affect reproductive function

Thyroid Function Tests	Both partners	Pre-IVF evaluation	To assess thyroid health	Identifies thyroid abnormalities that may affect fertility or pregnancy
Transvaginal Ultrasound	Female	Pre-IVF evaluation	To examine the ovaries and uterus	Assesses ovarian structure, follicle count, uterine anatomy, endometrium, fibroids, cysts, and other abnormalities
Hysteroscopy / Saline Scan	Female	Pre-IVF / Before embryo transfer, if indicated	To evaluate the uterine cavity in greater detail	Identifies uterine abnormalities that may affect embryo implantation
Semen Analysis	Male	Pre-IVF evaluation	To assess male reproductive potential	Measures sperm count, motility, and morphology
Male Hormonal Profile	Male	Pre-IVF, when indicated	To investigate possible hormonal causes of male infertility	Provides information about hormonal factors affecting sperm production and reproductive function
General Health Screening	Both partners	Pre-IVF	To assess overall health and treatment readiness	Identifies medical conditions or factors that may need attention before treatment
HIV Screening	Both partners	Pre-IVF	Infection screening and treatment safety	Identifies HIV infection and helps guide appropriate treatment and safety measures

Hepatitis B & C Testing	Both partners	Pre-IVF	Infection screening	Identifies hepatitis infection and associated treatment/pregnancy considerations
Syphilis Testing (VDRL/RPR)	Both partners	Pre-IVF	Infection screening	Identifies possible syphilis infection
Genetic Carrier Screening	Both partners, when indicated	Pre-IVF	To identify inherited genetic risks	Determines whether either partner may carry certain inherited conditions that could affect offspring
Karyotyping	Both partners, when indicated	Pre-IVF	To investigate chromosomal abnormalities	Evaluates chromosome number and structure; may be relevant in selected infertility or recurrent IVF failure cases
Endometrial Receptivity Assessment	Female, when indicated	Before embryo transfer	To evaluate uterine lining readiness	Provides information about endometrial receptivity and potential timing for implantation

What the evaluation is trying to establish

The overall pre-IVF assessment is essentially trying to answer four questions:

1. How is the ovarian reserve and function?

AMH, FSH, LH, ultrasound and follicle assessment provide information that can help the specialist understand ovarian reserve and plan stimulation.

2. Is there a male-factor issue?

Semen analysis provides the basic assessment of **count, motility and morphology**, with additional male testing when indicated.

3. Is the uterus suitable for implantation?

Ultrasound and, when necessary, hysteroscopy or saline-based evaluation look for structural or endometrial issues that could affect implantation.

4. Are there hormonal, infectious, genetic or general health factors that need consideration?

Blood tests and selected advanced investigations help identify conditions that may influence fertility treatment or pregnancy.

Important distinction

Not every test listed in the report is necessarily performed for **every IVF patient**.

There is a core pre-IVF evaluation—such as fertility history, hormonal assessment, ultrasound, infection screening and semen analysis—while tests such as **karyotyping, genetic screening, hysteroscopy, or endometrial receptivity testing may be performed only when clinically indicated**.

Summary

Stage: Before commencing IVF / treatment planning

Primary objective: Establish baseline reproductive health and identify factors that may influence IVF treatment and pregnancy

Female assessment: Ovarian reserve + hormones + ovarian/uterine anatomy

Male assessment: Semen quality + additional hormonal evaluation when required

Safety assessment: Infectious disease and general health screening

Advanced assessment: Genetic/chromosomal or uterine investigations in selected cases

Final use of information: Findings are used by the fertility team to determine the **appropriate IVF strategy, stimulation approach, fertilization method, and whether additional evaluation or treatment is required before proceeding**.

Pregnancy/antenatal stage reports

For the, the equivalent table should focus on tests, scans, and routine tracking that help the clinician monitor **maternal health, fetal development, pregnancy progression, and emerging complications**.

The exact schedule varies by country, clinic, maternal risk factors, and whether the pregnancy is singleton or multiple. The table below is a practical **IVF/pregnancy report-summary framework**, not a rigid clinical protocol. Routine antenatal care commonly includes blood pressure, urine testing, blood tests, ultrasound, fetal growth assessment, and screening at defined gestational ages. ([nhs.uk](https://www.nhs.uk))

Pregnancy / Antenatal Monitoring Table

Test / Evaluation	Applicable For	Stage / Timing	Why is it done?	What information does it reveal / Purpose
Pregnancy Confirmation – β-hCG	Female / Pregnancy	Early pregnancy, after embryo transfer or missed period	To confirm biochemical pregnancy	Confirms pregnancy and, when serially measured, helps assess early pregnancy progression
Early Pregnancy Ultrasound / Viability Scan	Female / Fetus	Early pregnancy, commonly around 6–8 weeks when indicated	To confirm that the pregnancy is developing in the uterus and assess early viability	Confirms gestational sac, location, fetal development/heartbeat when visible, and number of pregnancies
Blood Pressure Monitoring	Female	Throughout pregnancy; at routine antenatal visits	To monitor maternal cardiovascular status	Helps identify hypertension and possible preeclampsia ; repeated measurements are important for trend monitoring. (nhs.uk)

Urine Examination	Female	Throughout pregnancy / routine visits	To screen for urinary and pregnancy-related abnormalities	Can identify proteinuria, urinary infection indicators and, in context with BP, possible preeclampsia. (nhs.uk)
CBC / Hemoglobin	Female	Initial assessment and commonly repeated later in pregnancy	To assess maternal blood health	Detects anemia and provides information about hemoglobin and blood-cell status
Blood Group & Rh Typing	Female	Early pregnancy	To determine maternal blood group and Rh status	Identifies Rh-negative pregnancies where appropriate preventive management may be required; blood group is also important for pregnancy/delivery planning. (nhs.uk)
Infectious Disease Screening – HIV, Hepatitis B, Syphilis	Female; some screening may involve both partners depending on setting	Early pregnancy	To identify infections that may affect maternal or fetal health and guide management	Detects maternal infection and enables appropriate treatment/prevention measures. (nhs.uk)
Thyroid Function Tests	Female, when indicated / according to clinical protocol	Early pregnancy or when clinically indicated	To evaluate thyroid function	Identifies thyroid abnormalities that may require monitoring or treatment during pregnancy

Blood Glucose / Diabetes Screening	Female	Early pregnancy if risk factors; routine screening commonly around 24–28 weeks	To identify gestational diabetes	Detects abnormal glucose regulation and helps guide maternal/fetal management. (nhs.uk)
First-Trimester / Dating Ultrasound	Female / Fetus	Approximately 11–14 weeks	To establish/confirm gestational age and assess early fetal development	Provides dating measurements, confirms number of fetuses, and may assess early anatomy. (nhs.uk)
Nuchal Translucency (NT) / Combined Screening	Female / Fetus	Approximately 11–14 weeks , when selected	To screen for increased chance of certain chromosomal conditions	Combines ultrasound findings and blood markers to estimate the chance of conditions such as Down syndrome, Edwards syndrome and Patau syndrome. (nhs.uk)
NIPT / Cell-Free DNA Screening	Female / Fetus	From early pregnancy, according to test protocol	To screen for selected fetal chromosomal abnormalities	Provides a screening estimate for selected chromosomal conditions; it is a screening test, not a definitive diagnosis
Second-Trimester Serum Screening / Quad Test	Female / Fetus	Approximately 14–20 weeks , particularly when first-trimester combined screening was not performed	To screen for selected chromosomal/fetal conditions	Provides a calculated risk estimate for certain fetal conditions. (nhs.uk)

Anomaly / Detailed Anatomy Scan	Female / Fetus	Approximately 18–21 weeks	To assess detailed fetal anatomy and development	Evaluates fetal structures, growth, placenta and other anatomical findings; can identify signs of several structural abnormalities. (nhs.uk)
Fetal Echocardiography	Female / Fetus, when indicated	Usually around 20–22 weeks	To examine fetal cardiac structure and function in greater detail	Assesses fetal heart anatomy and identifies suspected cardiac abnormalities
Placental Assessment	Female / Fetus	During ultrasound examinations, particularly mid-pregnancy and later	To assess placental location and appearance	Identifies issues such as low-lying placenta/placenta previa and helps with later pregnancy and delivery planning. (nhs.uk)
Fetal Growth Ultrasound	Female / Fetus	Later pregnancy, frequency based on risk	To monitor fetal growth and development	Tracks fetal size/growth trajectory, amniotic fluid and other parameters; additional scans may be required in high-risk pregnancies
Fundal Height / Abdominal Growth Measurement	Female / Fetus	Generally from mid-pregnancy onward during routine visits	To monitor uterine/fetal growth over time	Provides a simple longitudinal indicator of fetal growth and can identify pregnancies requiring further assessment. (nhs.uk)

Fetal Movement Assessment	Female / Fetus	Mainly later pregnancy	To monitor perceived fetal activity	Changes in fetal movement can indicate a need for clinical assessment. Routine visits commonly include discussion of fetal movements. (nhs.uk)
Doppler Studies	Female / Fetus, when indicated	Usually later pregnancy / high-risk monitoring	To assess fetal and placental blood flow	Provides information about placental circulation and fetal blood-flow patterns; particularly useful when growth restriction or other complications are suspected
Non-Stress Test (NST)	Female / Fetus, when indicated	Usually later pregnancy / high-risk pregnancy	To assess fetal well-being	Monitors fetal heart-rate response and can help assess fetal status
Biophysical Profile (BPP)	Female / Fetus, when indicated	Usually later pregnancy / high-risk pregnancy	To assess fetal well-being using ultrasound and fetal heart-rate assessment	Combines parameters such as fetal movement, tone, breathing movements, amniotic fluid and NST findings
Repeat CBC / Maternal Blood Tests	Female	Commonly around the third trimester	To reassess maternal health	Monitors anemia and other blood abnormalities as pregnancy progresses
Group B Streptococcus (GBS) Screening	Female	Commonly 35–37 weeks in protocols that use universal screening	To determine whether the mother is colonized with GBS	Identifies maternal GBS colonization so appropriate intrapartum management can be planned. (CDC Stacks)

Fetal Presentation / Position Assessment	Female / Fetus	Particularly around 36 weeks onward	To determine fetal position before delivery	Identifies whether the baby is head-down, breech or in another presentation, supporting delivery planning. (nhs.uk)
Amniotic Fluid Assessment	Female / Fetus	During later pregnancy ultrasounds / when indicated	To assess fetal environment	Identifies abnormally low or high amniotic fluid, which may require additional monitoring
Late-Pregnancy Ultrasound / Well-being Assessment	Female / Fetus, when indicated	Third trimester	To assess fetal growth and well-being	Evaluates growth, fluid, placenta, fetal position and other parameters relevant to delivery planning

For an IVF-specific pregnancy report

For your **IVF Report Summary POC**, I would give particular importance to these fields when they appear in a pregnancy report:

Pregnancy status

- Gestational age
- EDD
- Singleton / twins / multiple pregnancy
- IVF/FET cycle details, if available
- Embryo transferred, if relevant

Maternal monitoring

- Blood pressure trends
- Weight, if documented
- Hemoglobin/CBC
- Blood group/Rh
- Glucose/OGTT
- Thyroid values
- Urine/protein findings

Fetal monitoring

- CRL / gestational measurements
- Fetal heart rate
- NT
- Anatomy/anomaly findings

- Estimated fetal weight
- Growth percentile/trend
- Amniotic fluid
- Placental location
- Fetal presentation
- Doppler findings, when performed

Screening

- NIPT / combined screening
- Genetic screening results
- Infection screening
- GBS status
- Any positive/abnormal screening requiring follow-up

High-value clinical flags

- Hypertension / preeclampsia indicators
- Gestational diabetes
- Fetal growth restriction
- Abnormal Doppler
- Placental abnormalities
- Abnormal fetal anatomy findings
- Multiple pregnancy complications
- Any investigation marked **abnormal / positive / high risk / requires follow-up**

This is the information that would make a pregnancy-report summary substantially more useful to an experienced IVF practitioner than simply summarizing the narrative text. Routine antenatal care is longitudinal, so **trends and changes between visits** are particularly valuable; for example, fetal growth, blood pressure, urine protein, fetal heart rate, and gestational age should ideally be summarized across reports rather than treated as isolated values. ([nhs.uk](https://www.nhs.uk))